ML Builder Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One complete expo evidence and review record.

Trace files,calculate supported metrics and answer critique without inventing absent artifacts.

Keep one habit: W1–W3/W5 → selected95%claim → actual project bundle → changed-case readiness check.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Train, Analyze, Improve. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Capstone Train, Analyze, Improve

> **My remembered idea, example and correction**

Model Cards and Human Review

> **My remembered idea, example and correction**

Improve Generalization

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main separate paper defense:12 fictional parcel references, blocked positive, candidateTN7/FP 1/FN 2/TP 2 gives 9/12 accuracy, 2/3 precision, 2/4 recall; fixed-clear baseline 8/12, precisionundefined, recall 0/4. Complete errorsR08 falsealarm, R09/R10 misses; no condition causes supplied. Exact training/code/split artifacts remain unprovided.

  Separate picture 15 indoor views from 5 plants×3:12 correctcandidate, 10 baseline; candidateF03 falsealarm, F11/F14 misses. No outdoor trial. Your own project must cite its actual records. Selected95%on20 claim has no auditable baseline, split history or errors.

- Draft a three-sentence opening with purpose, a result and a limit.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main separate paper defense:12 fictional parcel references, blocked positive, candidateTN7/FP 1/FN 2/TP 2 gives 9/12 accuracy, 2/3 precision, 2/4 recall; fixed-clear baseline 8/12, precisionundefined, recall 0/4. Complete errorsR08 falsealarm, R09/R10 misses; no condition causes supplied. Exact training/code/split artifacts remain unprovided.

  Separate picture 15 indoor views from 5 plants×3:12 correctcandidate, 10 baseline; candidateF03 falsealarm, F11/F14 misses. No outdoor trial. Your own project must cite its actual records. Selected95%on20 claim has no auditable baseline, split history or errors.

- Name the files needed to audit one reported accuracy figure.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main separate paper defense:12 fictional parcel references, blocked positive, candidateTN7/FP 1/FN 2/TP 2 gives 9/12 accuracy, 2/3 precision, 2/4 recall; fixed-clear baseline 8/12, precisionundefined, recall 0/4. Complete errorsR08 falsealarm, R09/R10 misses; no condition causes supplied. Exact training/code/split artifacts remain unprovided.

  Separate picture 15 indoor views from 5 plants×3:12 correctcandidate, 10 baseline; candidateF03 falsealarm, F11/F14 misses. No outdoor trial. Your own project must cite its actual records. Selected95%on20 claim has no auditable baseline, split history or errors.

- A peer asks whether your model works for every school. Give a supported response.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main separate paper defense:12 fictional parcel references, blocked positive, candidateTN7/FP 1/FN 2/TP 2 gives 9/12 accuracy, 2/3 precision, 2/4 recall; fixed-clear baseline 8/12, precisionundefined, recall 0/4. Complete errorsR08 falsealarm, R09/R10 misses; no condition causes supplied. Exact training/code/split artifacts remain unprovided.

  Separate picture 15 indoor views from 5 plants×3:12 correctcandidate, 10 baseline; candidateF03 falsealarm, F11/F14 misses. No outdoor trial. Your own project must cite its actual records. Selected95%on20 claim has no auditable baseline, split history or errors.

- Write one next investigation tied to an observed **limitation**, and identify what must stay held out.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- The selected independent defense reports 95%on20 cases but omits its baseline, split history and errors. Treat95%as a reported claim:if exact and all 20 scored, it would mean 19 correct, but no ledger verifies that. Request the missing evidence rather than invent a confusion matrix or assume reliability across schools. Keep this case separate from the worked 12-case defense and pictured 15-photo record.

- Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Deliver the final Responsible Plant Classifier product card, live **evidence** trace, model card, and human-oversight commitment.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Trace **evidence** for Machine learning project **defense**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for Machine learning project **defense**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Final classifier gets16/20 correct; two of four errors are false negatives.
- Report supported metrics and what is missing.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Your expo model succeeds on three demonstration plants. What **evidence** should accompany the demo and what **limitation** should you state?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 36 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Final counts are TP=12,FN=3,FP=5,TN=20. Compute accuracy, precision and recall, then state one **limitation** and a review trigger for the model **defense**.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Review a **defense** claiming 95 percent success on 20 cases but omitting the baseline, split history and errors. Give three revision requests and explain why polish is insufficient.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Present a plant-classifier expo claim that another team could check.
- Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct. Three candidate failures are recorded; no outdoor trial exists.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/36

ML Builder Expo

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show denominator

Candidate12/15=80%; baseline10/15≈66.7% on the same cases.



2. Show independence

Explain five held-out plant identities and three related views per plant.



3. Show failures

Present the three failure records and indoor-only limitation beside the headline rate.

CHANGE ONE THING

The poster omits the baseline and calls the model "field ready".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Present a plant-classifier expo claim that another team could check.
- Which labelled step matches this detail? Present the three failure records and indoor-only **limitation** beside the headline rate.
- 1. Show failures
- 2. Show independence
- 3. Show denominator
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Show failures Present the three failure records and indoor-only **limitation** beside the headline rate. Read the picture in order.

Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: The poster omits the baseline and calls the model “field ready”. Which conclusion is supported by the supplied facts?
- 1. The headline exceeds the supplied **evidence**.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The headline exceeds the supplied **evidence**. No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Repair and keep the record

Review Capstone Train, Analyze, Improve (ai-4/35). Goal: Run a frozen final evaluation with complete accounting. Short practice: Why is “the revision is better” incomplete?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, **reflection**s, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

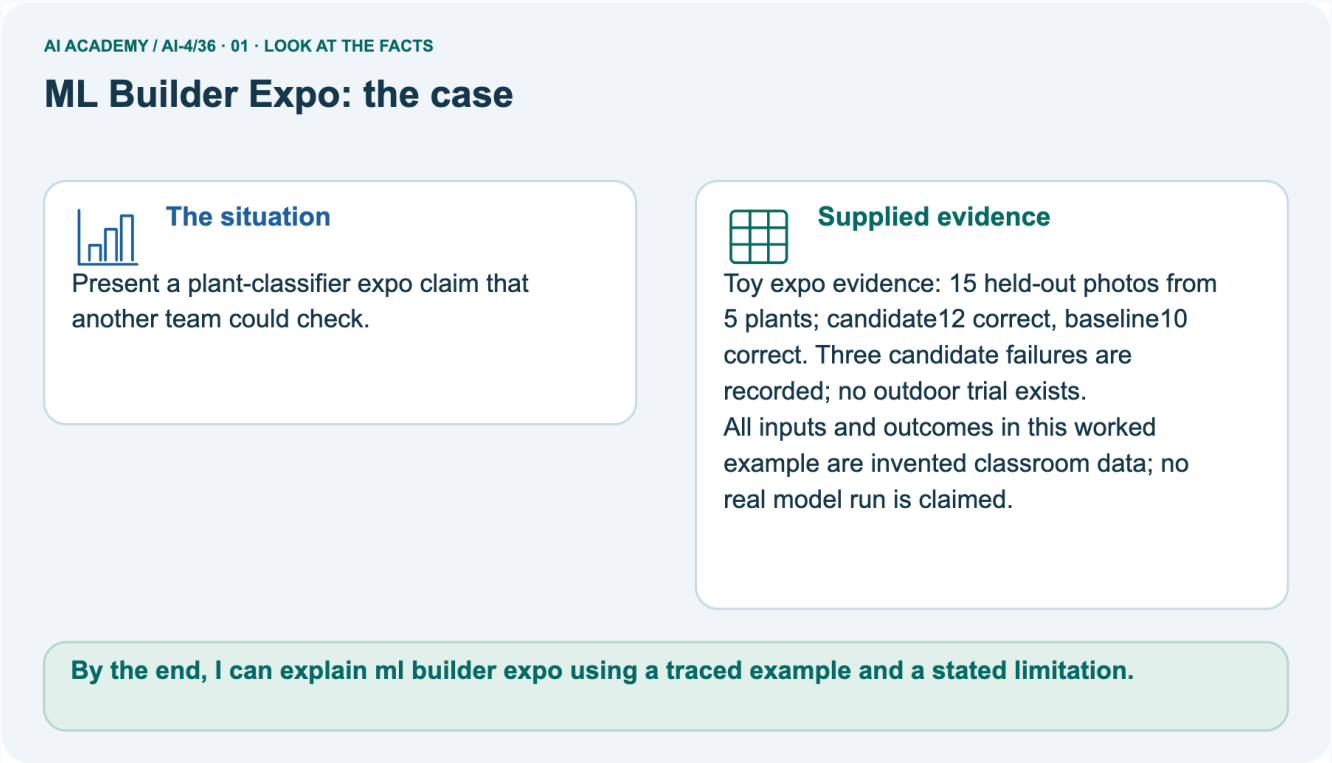
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. ML Builder Expo: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

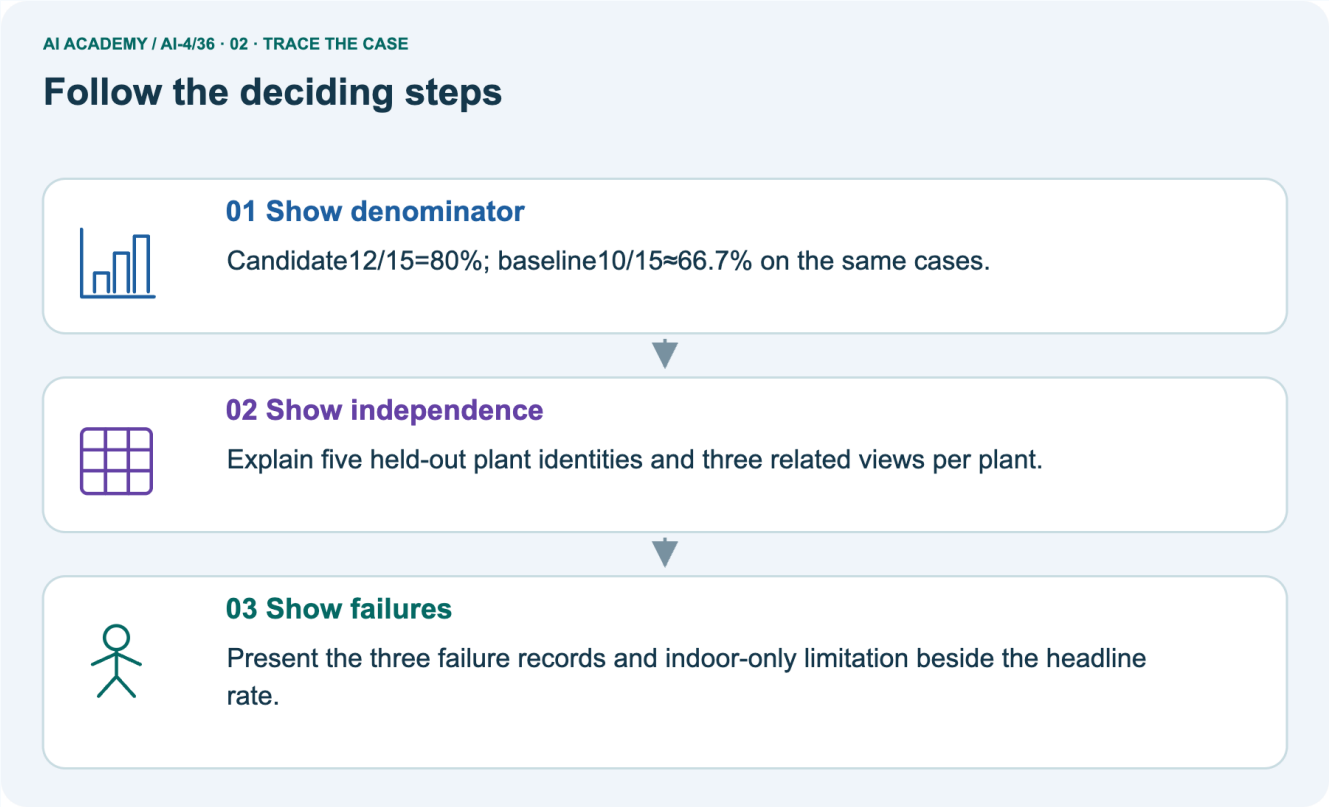
Present a plant-classifier expo claim that another team could check. Supplied facts: Toy expo **evidence**: 15 held-out photos from 5 plants; candidate12 correct, baseline10 correct.

Three candidate failures are recorded; no outdoor trial exists. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



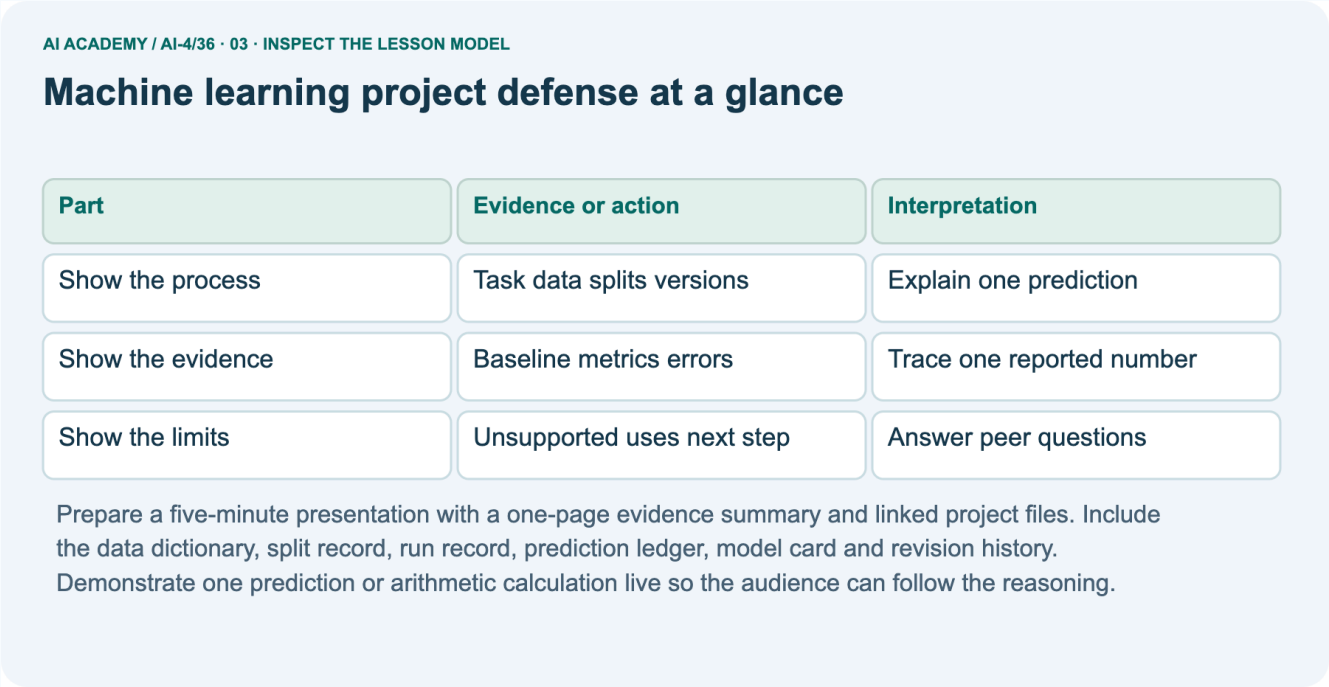
Follow the labelled picture with the matching explanation. Open the vector diagram

Show denominator: Candidate12/15=**80%**; baseline10/15≈66.7% on the same cases. → Show independence: Explain five held-out plant identities and three related views per plant. → Show failures: Present the three failure records and indoor-only **limitation** beside the headline rate.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Machine learning project defense at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

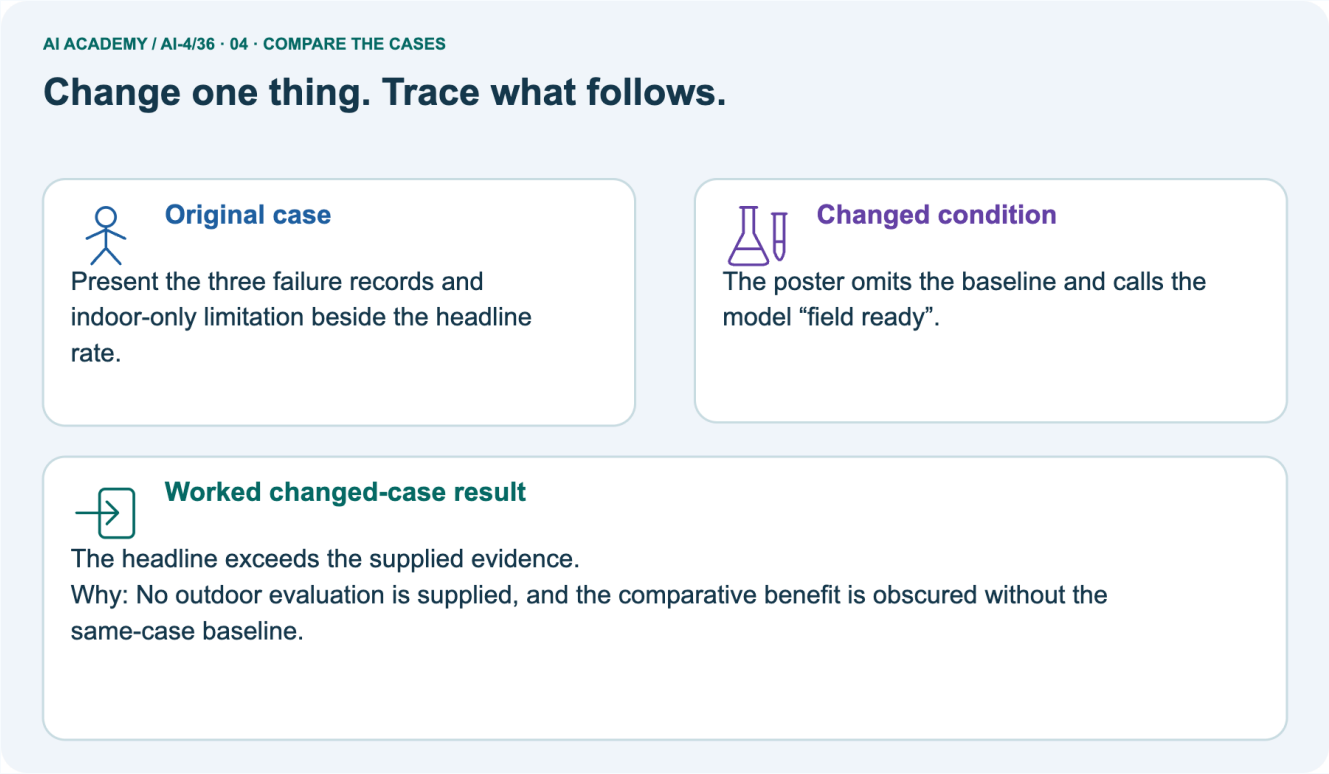
Machine learning project **defense** at a glance. Show the **process** | Task data splits versions | Explain one **prediction**. Show the **evidence** | Baseline metrics errors | Trace one reported number.

Show the limits | Unsupported uses next step | Answer peer questions

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

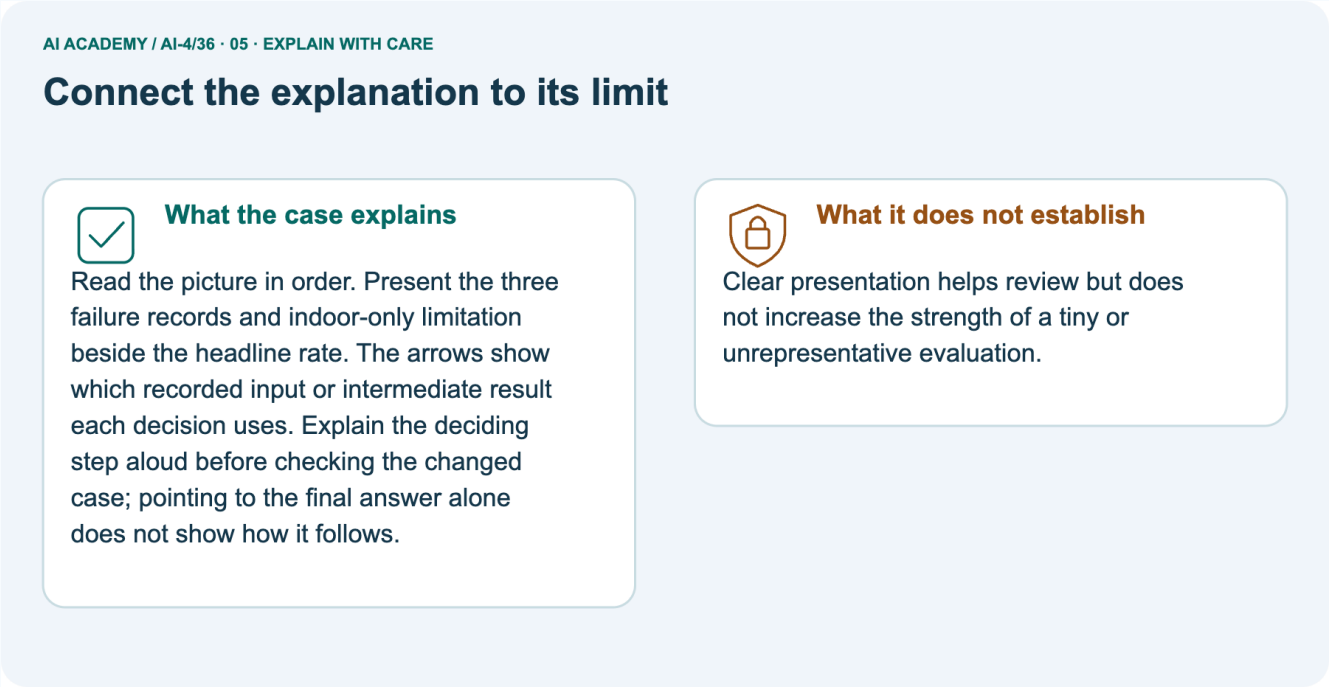
Original case: Present the three failure records and indoor-only **limitation** beside the headline rate. Changed condition: The poster omits the baseline and calls the model “field ready”.

Expected result: The headline exceeds the supplied **evidence**. Reason: No outdoor evaluation is supplied, and the comparative benefit is obscured without the same-case baseline.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Present the three failure records and indoor-only **limitation** beside the headline rate. The arrows show which recorded **input** or intermediate result each decision uses.

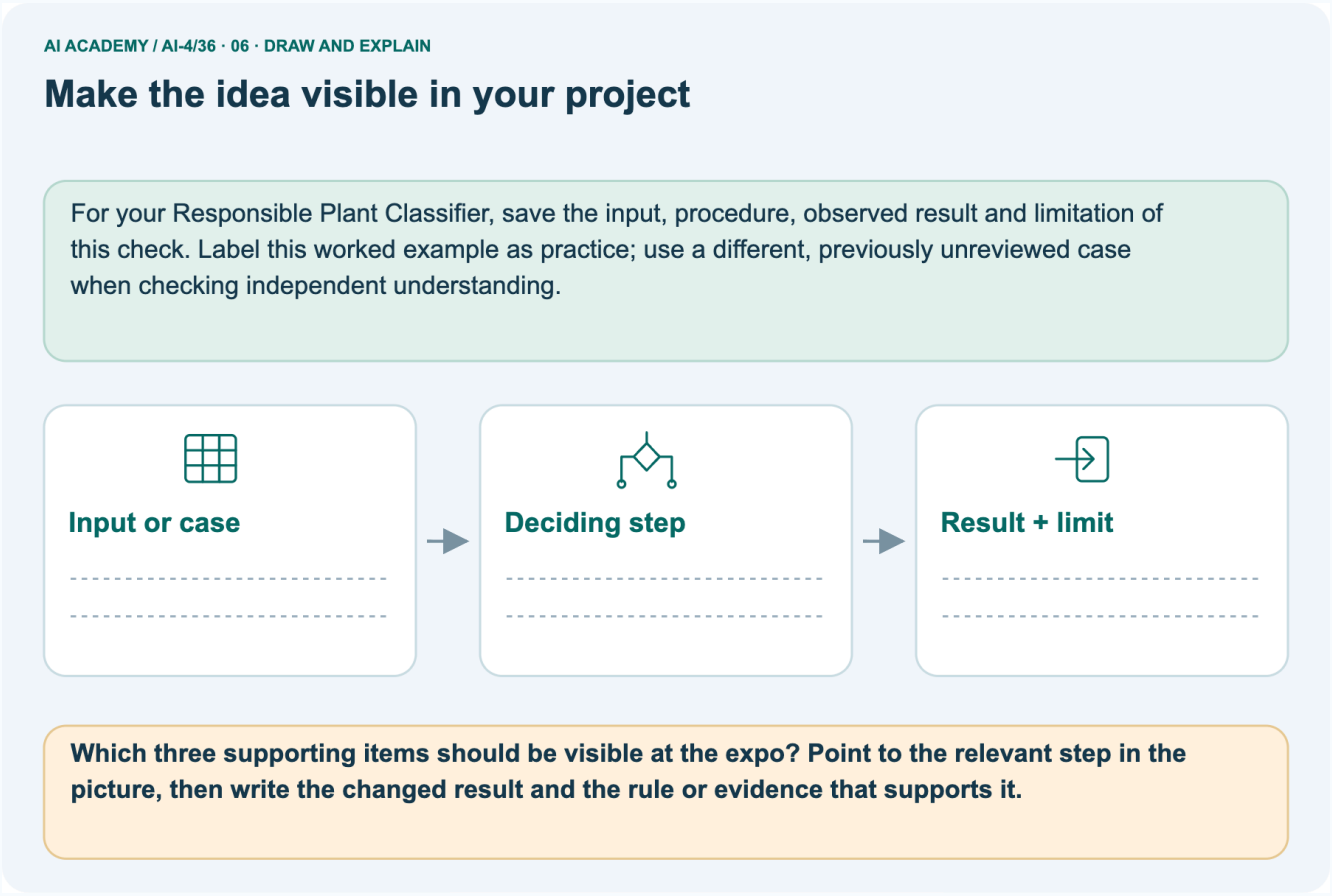
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Clear presentation helps review but does not increase the strength of a tiny or unrepresentative evaluation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which three supporting items should be visible at the expo? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. ML Builder Expo: the case

02. Follow the deciding steps

03. Machine learning project defense at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.